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Project Title:	HZE-induced mammary cancer development processes in murine and 'humanized' models, and their influence on radiation quality functions		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Task Description:	The proposed research used a model of breast cancer development in which mouse or human mammary stem cells are used to grow mammary glands in mouse mammary fat pads. When the stem cells used are irradiated prior to transplant, progression from very early stages of cancer (ductal dysplasias) can be monitored. We used this model to compare the development of cancer from cells exposed to gamma-rays versus cells exposed to HZE particles that are found in galactic cosmic rays. The histological progression of pre-cancerous lesions was monitored using markers of cell growth, senescence, apoptosis, proliferation, and angiogenesis. This information was to be correlated with global gene expression profiling of non-persistent and persisting dysplasias to gain further insight into the molecular mechanisms driving progress towards cancer after exposure to the different types of radiation. Studying the cancers that are initiated by radiation in both mouse and human mammary cells allows a direct comparison of cancer processes that may differ between species as well as between radiation types. This is important, as most assessments of human space radiation risk are based on experiments in rodents. In summary, this work was to assess the standard approach for generating HZE risk estimates in humans, which is to use gamma-ray to HZE quality functions that have been derived from rodent studies, and then to apply them to gamma-ray-based human epidemiological data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work was designed to provide insight into the cancer risks arising from exposure to space radiation. It asked questions about how cancer is started by radiation exposure, how cancer progresses, and how these processes might be the same and different after exposure to radiation encountered in space, and the more common gamma-rays encountered on Earth. Since breast cancer is one of the most common cancers associated with radiation exposure, this is an important model system. Understanding the progression of breast cancer after exposure to space radiation will help NASA to make better risk estimates for their long-term exploration missions. A better understanding of the factors that control breast cancer progression in general can also have a large impact on Earth, by suggesting better intervention or treatment strategies for this very common disease.

We made progress in addressing all three specific aims, despite a series of setbacks with our subcontract. The majority of the work was in support of specific aims 1 and 2.
The goal of Specific Aim 1 is “to quantitatively compare early tumor initiating events induced by HZE vs. gamma rays in murine mammary epithelial stem cells using the well-established ductal dysplasia model as well as a 3D in vitro model.” The experiments were designed to both quantify and characterize potential differences.

We have also used mammosphere cultures to examine effects of irradiation on the survival of stem cells in the irradiated mammary gland as measured by the frequency of mammospheres obtained from control and irradiated mice. Mammary cells were obtained one week post irradiation. We observed an approximately 50% reduction in the number of mammospheres obtained following gamma irradiation, which did not appear to be dose dependent over the 0 to 4 Gy dose range. A comparison of the effects of 1 Gy of gamma rays, protons, or iron showed a similar decrease that was not dependent on radiation type.

Work on this aim initially used the classic ductal dysplasia model to quantify initiated cells as measured by the frequency of persistent dysplasias in cleared mammary fat pads 16 weeks after injection of mammary epithelial cells. The mammary cells were obtained from 12 week old female BALB/c mice 16 weeks after irradiation with Cs-137 gamma rays, SPE 1972 protons, or 1 GeV/n Fe-56 ions. Following injection of cells, the frequency of persistent dysplasias was quantified 16 weeks later. This model was heavily dependent upon the ability to obtain a consistent number of mammary outgrowths from experiment to experiment in order to quantify dysplasias. Unfortunately, we found a wide variation in the ability to obtain fully developed mammary glands using this model. In some studies more than 90% of the mammary fat pads injected had outgrowths while in others as few as 50% yielded evaluable mammary outgrowths. As a result, data were quite variable and thus required us to perform many more experiments to obtain reliable results. The mice from these experiments also had a high rate of infection, necessitating more repeated experiments. During this time, we also devoted considerable effort to optimizing all steps of mammosphere shipping, RNA recovery, amplification and labeling.

Because of the variability encountered in the classic assay emerged, we compared three approaches for quantification of initiated cells and development of lesions in the murine transplantation model, and found that an intraductal transplantation model based on a recent paper by Behbod et al. (Breast Cancer Res, 11(5): R66, 2009), proved much more efficient than the initial cleared fat pad approach. We found that intraductal injection of murine mammary epithelial cells provides similar data to that previously obtained using the cleared fat pad dysplasia model (Adams et al. Cancer Res 47(16): 4425-4431, 1987) but requires less time, is more reproducible, and provides more tissue for analysis. Furthermore, based on Behbod et al. 2009, this approach should enable direct comparison of radiation effects in murine versus human breast epithelial cells.

This approach is not dependent upon the development of mammary glands in order to quantify the dysplastic nature of injected cells. The frequency of dysplastic outgrowths scored in the intraductal injection model tracked well with the frequency in the cleared fat pad approach in studies in which greater than 90% of mammary fat pads injected produced mammary outgrowths. Histological characterization of the outgrowths with respect to their dysplastic nature was performed, and approximately 90% of the outgrowths were classified as dysplastic while 10% could be classified as ductal carcinoma in situ. Both classifications support the interpretation that the cells from which the outgrowths were derived were preneoplastic in nature. This approach also provided a greater number of dysplastic cells for analysis than the classic approach, as the outgrowths were confined in a well-defined area.

We found that the number of persistent dysplastic growths (fully initiated cells) is higher in mice transplanted with mammospheres derived from mice irradiated with Fe particles compared to those irradiated with gamma rays. This supports our hypothesis, that in comparison to low LET gamma-rays, HZE particle exposure would produce more efficient conversion of mammary cells to a pre-neoplastic state, similar to the response previously seen for high LET neutron exposures. Once the new approach had been worked out, mammary epithelial tissue was harvested after exposure of Balb/c mice to SPE1972 Protons, 1 GeV/n 56Fe, 350 MeV/n 28Si, gamma rays, or from controls. Cells from the tissue were dissociated and grown in attachment-free and non-differentiating conditions to allow mammosphere formation. Mammospheres were collected, dissociated into individual cells, and cultured as secondary mammospheres, which were then collected and injected into recipient mice for the formation of dysplastic lesions. Samples of both the primary and secondary mammosphere cultures were retained for gene expression studies.

Task Progress:	While the first set of mice sacrificed (4 weeks) showed similar lesion frequencies to those that had been seen previously, at the second set of sacrifices (12 weeks) we found that all of the initial outgrowths had regressed when allowed to remain in situ. The reason for this discrepancy from previous results is not known, and when the entire experiment was repeated, no lesions were observed, so the unexpected regression issue could not be resolved. The goal of Specific Aim 2 was “to assay both phenotypic progression and molecular profiles using the mammary ductal dysplasia and 3D stem cell models to determine whether HZE irradiation results in qualitatively different processes, compared with gamma-ray irradiation, that point to different mechanisms of carcinogenesis.” We compared expression of genes considered putative stem cell markers (Klf4, Nanog, Wnt1) and epithelial cell markers (cytokeratin 14 - basal myoepithelial, cytokeratin 18 - luminal) in the primary and secondary mammospheres used for injection. We found that the stem cell markers Klf4 and Wnt1, as well as the basal myoepithelial marker cytokeratin 14 were expressed at higher levels in secondary compared to primary mammospheres, suggesting a shift in cell types and a possible enrichment for stem cells with the second round of culture. We also looked at the potential effect of radiation exposure to the donor mice on expression of these genes in primary and secondary mammospheres. All the genes examined were up-regulated in primary mammospheres from irradiated mice, but in the secondary mammospheres expression levels were beginning to return to the levels in unexposed controls. This may be a function of the additional ten days between irradiation of the animals and the measurement of gene expression in the secondary mammospheres, reflecting a gradual reduction in the radiation effect as time progresses. It may be that gene expression is not the most appropriate method for assessing relative stem cell enrichment, however. RNA was harvested from lesions from mice sacrificed at 4 weeks. As noted above, for unknown reasons no lesions were identified in mice sacrificed at 12 weeks. Unfortunately, the RNA from the early lesions was of lower than expected quality, with RIN values between 4 and 6, indicative of degradation. In light of the low quality and missing late lesion samples, array hybridization was not undertaken. Due to the persistent technical difficulties encountered with the mouse lesions, we extended our work to include studies of human cancer stem cells in support of our aims. Cancer stem cells (CSC) can be identified in several different tumors based on the presence or absence of specific cell surface markers. Breast CSCs have been identified by the surface marker expression of CD44+/CD24low. CD44 is expressed on the surface of cancer stem cells from a variety of tumor types and high CD44 expression is associated with increased risk of metastasis and poor overall patient prognosis. CD44 positive breast cancer stem cells have been shown to be highly refractory to radiation and chemotherapeutics and are enriched following radiation treatment. We exposed exponentially growing cultures of breast cancer cell lines (MDA-231, MDA-468, and MCF-7) to gamma radiation and analyzed CD44 expression 24 and 5 days post-exposure using flow cytometry. The standard CD44 form (CD44s) was measured and a change in CD44s intensity was observed in the cell lines 24 and 5 days after exposure to gamma radiation. In MDA-231 the pattern of CD44s intensity was decreased at a dose as low as 0.1 Gy and increased following 10 Gy radiation exposure as early as 24 hours and persisted after 5 days. MDA-231 and MDA-468 displayed similar CD44s patterns 5 days post-exposure. Interestingly, MCF7 showed decreased CD44s levels for each radiation dose at 24 hours but no change at 5 days post-exposure. The patterns of CD44s expression were stable for at least two weeks in progeny of irradiated cells at all doses studied. Taken together these data show that radiation alters CD44 signaling in breast cancer cells in a dose dependent manner. Mammalian tissues generally experience oxygen (O2) concentrations in the range of 2 to 8%, far lower than the standard atmospheric O2 environment (20%) in which most cells are cultured in the laboratory. When grown under atmospheric O2 many human cell lines have been shown to accumulate oxidative damage and undergo premature senescence. To test the effect of O2 culture conditions on the intensity of CD44s staining we cultured MDA-231, MDA-468, and MCF7 at 3% O2. The results at 24 hours post exposure revealed a different pattern of CD44s intensity compared to 20% O2 in each cell line. Radiation dose dependent differences were not observed in MDA-231 or MCF7. In MDA-468 increased CD44s intensity was observed after exposure to 10 Gy. At 5 days after radiation exposure the pattern of CD44s in MDA-231 was similar when grown under 3% or 20%. In MDA-468 cells grown at 3% O2 CD44s intensity decreased (relative to control) for both the 0.1 and 1.0 Gy samples. Although no radiation effect on CD44s intensity was observed in MCF7 cells grown at 20% O2, under 3% O2 they showed a CD44s pattern similar to MDA-231 and MDA-468. The data confirm our initial finding that low dose exposures seemed to decrease the “stemness” of cancer cell populations as defined by CD44 expression, while high doses increase the tendency to “stemness.” It also indicates that O2 tension influences cancer stem cell physiology and radiation response, and should be considered when using in vivo models. We also completed the analysis and publication of some of our earlier transcriptomic work that was based on NSRL irradiation of cell lines. Having this information in hand will help to provide a baseline for comparisons of radiation responses in other model systems relevant to space radiation risk.
Bibliography Type:	Description: (Last Updated: 09/17/2020)
Articles in Other Journals or Periodicals	Meador JA, Amundson SA. "Histone Gene Expression in Irradiated Cells." Center for Radiological Research 2009 Annual Report, New York, NY. p. 52-54. , Jun-2010
Articles in Other Journals or Periodicals	Meador JA, Ghandhi SA, Amundson SA. "Radiation Induced Changes in CD44 Expression and Hyaluronic Acid Synthesis in Breast Cancer Cells." Center for Radiological Research 2011 Annual Report, New York, NY. p. 36-38. , Jun-2012